VEIL: Variance-Echoed Instance-Less Prompt Learning for Membership-Resistant Federated Vision–Language Models

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Abstract

Federated prompt learning adapts frozen vision–language models through compact updates, yet those updates can preserve instance-specific membership traces. We introduce VEIL (Variance-Echoed Instance-Less Prompt Learning), a three-stage client mechanism that converts private embeddings into local class-distribution surrogates before prompt optimization and sanitizes only the resulting prompt delta. Its central operation samples covariance-shaped echoes around class prototypes without forming or communicating a full covariance matrix. We audit six complementary membership attacks at the stringent true-positive rate at 1% false-positive rate (TPR@1%FPR) using label-matched candidates, isolated audit randomness, three datasets, and repeated non-IID partitions. Against FedAvg defenses, VEIL attains the lowest cross-dataset mean attack leakage (0.0145) while matching Prompt-DP mean accuracy within 0.12 percentage points. Under the same six attacks, it also exceeds DP-FPL and FedASK accuracy on every dataset and reduces their cross-dataset mean leakage by 3.8% and 36.7%, respectively. DP-FPL remains better on DTD and slightly better in cross-dataset worst-attack leakage, exposing a genuine transfer boundary rather than universal dominance. VEIL is therefore an empirical membership defense, not a differential-privacy guarantee.

1 Introduction

Vision–language models such as CLIP provide transferable representations that can be adapted with a small set of learned context tokens rather than full-model fine-tuning (Radford et al. 2021; Zhou et al. 2022). Federated prompt learning combines this parameter efficiency with decentralized optimization: clients retain their images and communicate only prompt updates (McMahan et al. 2017; Zhao et al. 2023; Guo et al. 2024). The resulting protocol is cheap, but it exposes a recurring sequence of low-dimensional, task-specific updates. Such updates and the released prompt can leak whether a candidate example was used for local training, even when the frozen backbone never leaves the server.

Membership inference is particularly consequential at low false-positive rates, where an attacker can make high-confidence claims about a small number of people or records (Shokri et al. 2017; Carlini et al. 2022). Existing defenses offer incomplete choices. Differentially private optimization gives a formal guarantee, but noise can destabilize prompt learning in strongly non-IID, few-shot regimes (Abadi et al. 2016; Duan et al. 2023). Output-centered methods such as HAMP reduce overconfidence, yet do not directly erase update alignment (Chen and Pattabiraman 2024). Private federated prompt and low-rank training mechanisms, including DP-FPL and FedASK (also called FedSAK in some project documentation), change the communicated parameterization but remain subject to empirical auditing (Tran et al. 2025; Wen et al. 2025).

We ask whether the client can preserve the class structure needed for prompt learning without repeatedly optimizing on identifiable instance directions. Our answer is VEIL. Like a veil that preserves a silhouette while hiding identity, the client presents the prompt optimizer with variance-shaped echoes of its class distribution: the semantic outline remains recognizable, but the individual that generated it is obscured. The method operates entirely in the frozen image-embedding space. It compresses local class geometry into an implicit low-rank sampler, learns the prompt from distributional surrogates rather than raw instance anchors, and releases a norm-controlled prompt delta. These three stages target the representation and protocol channels without changing the frozen backbone or server optimizer.

Our contributions are threefold:

- We identify direct instance anchoring as a membership channel in federated soft-prompt tuning and derive an implicit class-geometry sampler that retains local semantic variation without communicating client statistics.
- We develop the three-stage VEIL mechanism: private geometry estimation, instance-obscured surrogate learning, and structured prompt release, while keeping CLIP frozen and aggregating only trainable parameters.
- We provide a protocol-aware benchmark that compares VEIL, DP-FPL, and FedASK under six attacks with exactly label-matched candidates and reports both per-attack leakage and utility across three non-IID datasets.

2 Related Work

Federated prompt learning. Prompt tuning adapts a pretrained model through a small trainable context (Lester, Al-

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Rfou, and Constant 2021; Zhou et al. 2022). FedAvg supplies the standard weighted aggregation rule for decentralized optimization (McMahan et al. 2017), and FedPrompt and PromptFL demonstrate the communication advantages of exchanging prompts (Zhao et al. 2023; Guo et al. 2024). DP-FPL factorizes global and personalized prompts and perturbs private updates, while FedASK aggregates both low-rank factors through an asymmetric double-sketch mechanism (Tran et al. 2025; Wen et al. 2025). Our work does not replace these mechanisms; it studies a complementary defense at the client representation and upload levels.

Membership inference and auditing. Black-box membership inference exploits differences in model behavior on training and non-training examples (Shokri et al. 2017). Nasr et al. extend the threat to passive and active white-box adversaries (Nasr, Shokri, and Houmansadr 2019). Modern attacks emphasize calibrated likelihood ratios and low-FPR evaluation (Carlini et al. 2022; Zarifzadeh, Liu, and Shokri 2024), while scalable quantile regression learns input-dependent non-member thresholds (Bertran et al. 2023). FedMIA uses other clients as a per-round null population and measures confidence or gradient–update alignment (Zhu et al. 2025). We retain each attack's information source but specialize features to frozen-backbone prompt tuning.

Membership defenses. DP-SGD clips per-example gradients and adds Gaussian noise (Abadi et al. 2016); private prompt learning applies related ideas to a much smaller parameter space (Duan et al. 2023). HAMP combines high-entropy soft targets, entropy regularization, and low-confidence prediction mapping (Chen and Pattabiraman 2024). In contrast, VEIL intervenes before the prompt loss sees an individual image embedding. Its geometry sampling resembles distributional feature augmentation, but all class statistics remain local and the objective is membership resistance rather than data sharing or domain generalization.

3 Motivation: Compact Updates Still Carry Individual Traces

Freezing CLIP eliminates backbone-gradient leakage but does not eliminate three prompt-specific membership channels. First, the loss and confidence trajectory of a member can remain systematically different over rounds. Second, the gradient induced by a candidate can align with the observed client prompt delta. Third, a final prompt may retain an input-dependent confidence gap. Because the trainable prompt is small, each private mini-batch contributes to the same compact parameter block; parameter efficiency can therefore concentrate rather than remove membership evidence.

Gradient perturbation alone is ill-suited to the few-shot non-IID regime. A local class may contain only a handful of examples, so indiscriminate noise can remove useful directions and create large run-to-run utility variation.

Our design follows a class-level sufficiency hypothesis. Prompt learning needs semantic position and local variation, whereas repeatable instance directions form an avoidable membership channel. This yields three requirements: preserve local geometry without exporting statistics, remove raw member features from the prompt objective, and control the residual update before aggregation. All three must preserve the frozen backbone and standard FedAvg server.

4 Problem Formulation and Threat Model

There are K clients. Client k owns $D_k = \{(x_i, y_i)\}_{i=1}^{n_k}$ and the server maintains trainable prompt θ^t. The CLIP image encoder ϕ and text encoder ψ are frozen. Given class c, $\psi(c; \theta)$ denotes its normalized text representation. Only tensors whose `requires_grad` flag is true are optimized and aggregated. Under FedAvg, the server computes

$$\theta^{t+1} = \sum_{k \in S_t} \frac{n_k}{\sum_{j \in S_t} n_j} \theta_k^{t+1}. \quad (1)$$

The adversarial server observes protocol messages from all selected clients and released prompt checkpoints, and can query the released model. It does not see raw client images, local feature banks, class means, or covariance factors. For a candidate (x, y) and target client k^*, it predicts whether $(x, y) \in D_{k^*}$. We evaluate at $\mathrm{TPR@1\%FPR}$, following low-FPR membership auditing practice (Carlini et al. 2022). VEIL is an empirical defense: its Gaussian upload perturbation is not claimed to satisfy a calibrated (ϵ, δ)-DP bound.

5 VEIL

VEIL separates the class information needed for adaptation from the instance directions that enable membership inference. "Instance-less" refers to the optimization interface: private examples estimate local geometry, but their embeddings never directly supervise the prompt. The complete mechanism contains the three stages in Algorithm 1.

5.1 Stage I: Private Semantic Geometry

Stage I summarizes each local class without exporting any statistic. Client k obtains the detached normalized feature $z_i = \phi(x_i)/\left\|\phi(x_i)\right\|_2$ from the frozen image encoder and computes the class center $\mu_{k,c} = n_{k,c}^{-1} \sum_{i:y_i=c} z_i$. Thus, subsequent prompt updates can depend on a class distribution rather than a named training point.

The centered factor preserves within-class directions without constructing a dense covariance. For $n_{k,c} \geq 2$, VEIL stores

$$G_{k,c} = \frac{1}{\sqrt{n_{k,c} - 1}} \begin{bmatrix} (z_1 - \mu_{k,c})^\top \\ \vdots \\ (z_{n_{k,c}} - \mu_{k,c})^\top \end{bmatrix}. \quad (2)$$

Its Gram matrix $G_{k,c}^\top G_{k,c}$ equals the empirical feature covariance, so multiplying by a Gaussian vector samples the observed class subspace without an eigendecomposition. A class with fewer than two examples uses isotropic noise with standard deviation 0.02.

All geometry remains inside the client. Neither the detached feature bank nor $\mu_{k,c}$, $G_{k,c}$, or any derived covariance is communicated.

Algorithm 1 Three-Stage VEIL Client Update

Require: Frozen image encoder ϕ, prompt θ^t, local data D_k
1: **Geometry:** build detached $(\mu_{k,c}, G_{k,c})$ using Eq. (2)
2: **Surrogates:** optimize only θ on prototypes and echoes using Eqs. (3)–(4)
3: **Release:** upload detached $\theta^t + \widehat{\Delta}_k^t$ using Eq. (5)

5.2 Stage II: Instance-Obscured Surrogate Learning

Stage II removes direct member anchors from prompt optimization. Each batch label is represented by the normalized class prototype $p_{k,c} = \mu_{k,c} / \|\mu_{k,c}\|_2$, rather than by the feature of the corresponding image.

Geometric echoes restore the diversity lost by prototype replacement. For a fresh standard Gaussian vector $\varepsilon \in \mathbb{R}^{n_{k,c}}$, VEIL draws

$$\widetilde{z}_{k,c} = \frac{\mu_{k,c} + s\varepsilon^\top G_{k,c}}{\|\mu_{k,c} + s\varepsilon^\top G_{k,c}\|_2}. \tag{3}$$

The perturbation follows the empirical class subspace, while its center is the class mean instead of an individual member. We use three echoes and geometry scale $s = 0.6$.

Prompt learning operates exclusively on the surrogate set. Let $\widetilde{B}_k$ contain one prototype and the echoes for each batch label. Only the text prompt remains differentiable, and the client minimizes

$$\mathcal{L}_k(\theta) = \frac{1}{|\widetilde{B}_k|} \sum_{(z,y)\in\widetilde{B}_k} -\log \frac{\exp\big(\tau z^\top \psi(y;\theta)\big)}{\sum_{c'} \exp(\tau z^\top \psi(c';\theta))}. \tag{4}$$

This objective retains discriminative class structure without repeatedly aligning the prompt to a private feature direction.

5.3 Stage III: Structured Prompt Release

Stage III limits the residual membership trace in the communicated prompt delta. For $\Delta_k^t = \theta_k^{t+1} - \theta^t$, the client applies one global norm bound and releases

$$\widehat{\Delta}_k^t = \Delta_k^t \min\left(1, \frac{C}{\|\Delta_k^t\|_2}\right) + \xi_k^t. \tag{5}$$

The noise satisfies $\xi_k^t \sim \mathcal{N}(0, (\rho C)^2 I)$; we use $C = 0.5$ and $\rho = 0.11$. Every released prompt is a detached copy, preventing graph or storage aliasing across clients and rounds.

The server interface is unchanged. It aggregates only the released trainable deltas with standard weighted FedAvg, while all CLIP backbone parameters remain frozen. A fixed temperature of four calibrates final-query logits, but it is not treated as a separate method stage because positive scaling does not alter the reported rank-based metrics.

VEIL leaves communication and server state unchanged. Its only additional state is the local detached feature bank, which costs $O(n_k d)$ memory for embedding dimension d.

The additional computation is confined to the client. One frozen-encoder pass constructs the bank, and echo generation costs $O(mn_b d)$ per batch of size n_b. No class statistic or auxiliary message crosses the client boundary.

Dataset	Classes	Train	Test	Mean/client
Flowers102	102	1,632	6,149	163.2
Caltech101	101	1,616	5,647	161.6
DTD	47	752	1,880	75.2

Table 1: Dataset scale under the common 16-shot protocol. Client counts vary under the non-IID partition; the last column reports their arithmetic mean.

6 Experimental Methodology

6.1 Research Questions

The evaluation answers three focused questions. **RQ1** asks whether VEIL improves attack-specific leakage under a common FedAvg backbone. **RQ2** compares VEIL with DP-FPL and FedASK/FedSAK under the same six attacks. **RQ3** measures the contribution of each VEIL stage.

6.2 Datasets and Federated Protocol

The benchmark covers three complementary visual domains. Flowers102 (Nilsback and Zisserman 2008), Caltech101 (Fei-Fei, Fergus, and Perona 2004), and DTD (Cimpoi et al. 2014) represent fine-grained appearance, heterogeneous objects, and material texture. We use 16 labeled training examples per class and the complete held-out test partition.

All methods share one non-IID federated protocol. Ten clients receive a Dirichlet partition with concentration $\alpha = 0.1$ and participate in five global rounds with one local epoch per round. The batch size is 16, the learning rate is 10^{-3}, and the prompt contains 16 context tokens.

The backbone and execution policy are fixed across comparisons. CLIP ViT-B/32 is loaded from local storage, its encoders remain frozen, and formal runs require CUDA. Reported means and sample standard deviations use three independent repetitions; the main matrix contains 54 runs, and the focused component study contains the full method plus three stage ablations.

Hyperparameters are selected once and then frozen across datasets. We choose the lowest worst-attack leakage among configurations that maintain at least 0.60 development accuracy. The artifact records the full search and randomness controls.

6.3 Baselines

The first comparison isolates defenses on the same FedAvg backbone. It includes no defense, Prompt-DP, HAMP, and VEIL, with method-specific settings listed in the artifact.

The second comparison changes the private training mechanism. It evaluates VEIL, DP-FPL, and FedASK (called FedSAK in some project documentation) using configurations aligned with their public repositories while fixing the data, optimization budget, and audit.

Each baseline retains its defining state. Prompt-DP perturbs only trainable prompt tensors, HAMP uses its high-entropy objective with deterministic argmax-preserving release, and DP-FPL keeps its client-local residual. DP-FPL accuracy is therefore evaluated with each client's global-plus-local prompt; the other mechanisms release one shared prompt.

6.4 Attacks and Metrics

The attack suite spans six complementary membership signals. We evaluate FedMIA loss, FedMIA update cosine, their fixed joint score, Nasr-style passive white-box inference, offline RMIA, and quantile-regression MIA. These attacks cover confidence, update alignment, prompt statistics, reference likelihoods, and input-dependent thresholds.

The audit prevents label-distribution shortcuts. Each run uses 64 members and 64 non-members with exactly matched label histograms, and audit loading has an independent random generator. Non-members are always outside the target client's training set.

The primary privacy metric is $\mathrm{TPR@1\%FPR}$, where lower is better. Clean accuracy is reported under no attack, and attack tables report the mean TPR across the three repetitions. Because 64 non-members make the 1% operating point an empirical zero-false-positive threshold, values are quantized and exact zeros should be interpreted cautiously.

7 Results

7.1 FedAvg Defense Comparison

Table 2 answers RQ1 by exposing every attack rather than only a worst-case or average summary. The clean column is measured without an attack, and the remaining columns report attack TPR on the same trained model.

VEIL is the most consistent defense on Flowers102 and Caltech101. It is best or tied among defenses on four attacks for Flowers102 and five attacks for Caltech101, while achieving the highest defended accuracy on Flowers102 and remaining within 0.7 points of the best defended Caltech101 accuracy.

DTD exposes the transfer limit of the fixed operating point. Prompt-DP is best on five attack columns, whereas VEIL is best only on the loss attack and loses 9.1 clean-accuracy points relative to no defense. This reversal motivates reporting the complete attack vector instead of one aggregate score.

7.2 Comparison with DP-FPL and FedASK under Six Attacks

Table 3 answers RQ2 with the same clean-and-attack layout as Table 2. Figure 1 provides a low-density visual summary; exact values remain in the table.

VEIL provides the clearest utility advantage. Its clean accuracy exceeds DP-FPL and FedASK on every dataset, averaging 60.0% versus 55.1% and 48.9%.

The Flowers102 attack profile shows the strongest privacy advantage. VEIL is best or tied on five attacks and reduces Nasr TPR from 7.3%/6.2% to 1.0%. On Caltech101, it is best on cosine, joint, and RMIA while adding 6.1 accuracy points over DP-FPL.

DTD remains the principal limitation. DP-FPL is best on five attack columns, while VEIL is best only on the joint attack. Across all datasets and attacks, VEIL still reduces mean leakage by 3.8% relative to DP-FPL and 36.7% relative to FedASK.

Formal accounting remains conservative. Without claiming client or example subsampling amplification, Gaussian-RDP upper bounds span $\epsilon = 46.51$–511.51 at $\delta = 10^{-5}$ across the selected DP baselines; these values do not establish strong formal privacy.

7.3 Three-Stage Ablation

Table 4 answers RQ3 with one dependency-aware ablation per stage. Removing Stage I suppresses geometric echoes, removing Stage II restores individual anchors, and removing Stage III disables update clipping and noise.

The full method is the only configuration that reaches the minimum on every attack. Removing Stage I introduces RMIA leakage, and removing Stage II further exposes Quantile MIA. Removing Stage III raises Nasr from 3.1% to 6.2% and Quantile MIA from 0% to 3.1%, although clean accuracy increases by 2.9 points. The result identifies structured release as the strongest privacy–utility lever and shows that the three stages are complementary.

8 Limitations

The privacy conclusion is empirical rather than formal. A low measured $\mathrm{TPR@1\%FPR}$ can reflect finite candidate resolution, and no finite attack suite proves the absence of leakage.

The experimental scope limits external validity. We study one frozen CLIP architecture, full participation, and three image datasets; larger backbones, partial participation, other modalities, and adaptive attackers remain open. Prototype replacement may also under-represent rare within-class modes.

DTD exposes a concrete transfer limitation. The fixed Flowers102-selected configuration loses accuracy without improving aggregate leakage, motivating cross-dataset adaptation of the geometry and release scales.

9 Ethical Statement

The intended benefit is reducing disclosure of whether a person's example was used during collaborative adaptation. The same mechanism could conceal data provenance or complicate accountability if deployed without audit logs. We therefore recommend reporting both empirical attack results and any formal privacy accounting, preserving model/data documentation, and avoiding claims that federated learning alone guarantees privacy.

10 Conclusion

We presented VEIL, a three-stage defense for federated CLIP prompt learning: estimate private semantic geometry, optimize on instance-obscured surrogates, and release a norm-controlled prompt delta. This interface removes individual embeddings from the prompt loss without communicating class statistics or changing the FedAvg server. Under six label-matched attacks, VEIL provides the strongest cross-dataset accuracy and mean-leakage averages among VEIL, DP-FPL, and FedASK, while DP-FPL remains preferable on DTD privacy and slightly better in worst-attack leakage

Dataset	Defense	No attack	Attack TPR@1%FPR ↓					
		Acc. ↑	Loss	Cos.	Joint	Nasr	RMIA	Quant.
Flowers102	No defense	66.1	0.5	0.5	1.0	4.2	2.6	2.6
	Prompt-DP	61.7	0.5	1.6	1.6	4.2	**0.0**	3.6
	HAMP	54.7	2.6	1.6	**0.5**	**0.0**	0.5	**2.1**
	VEIL	**62.0**	**0.0**	**0.5**	**0.5**	1.0	**0.0**	4.7
Caltech101	No defense	86.8	2.1	0.0	1.6	0.0	0.0	5.2
	Prompt-DP	**81.1**	3.1	3.1	2.6	5.2	**0.0**	**2.6**
	HAMP	79.7	**1.0**	3.6	4.2	**1.0**	0.5	8.9
	VEIL	80.4	**1.0**	**2.1**	**0.5**	**1.0**	**0.0**	3.1
DTD	No defense	46.7	1.6	1.6	1.0	1.0	0.5	2.1
	Prompt-DP	37.6	1.0	**1.6**	**0.0**	**0.0**	**0.0**	**0.0**
	HAMP	**43.2**	2.6	2.1	1.0	1.0	0.5	0.5
	VEIL	37.6	**0.5**	3.1	1.0	2.1	3.6	1.0

Table 2: FedAvg defense comparison. All values are percentages averaged over three repetitions. The no-defense row is a reference; bold marks the best value among the three defenses, including ties. Shaded rows denote our method.

Dataset	Method	No attack	Attack TPR@1%FPR ↓					
		Acc. ↑	Loss	Cos.	Joint	Nasr	RMIA	Quant.
Flowers102	DP-FPL	55.5	**0.0**	2.6	2.6	7.3	**0.0**	**3.1**
	FedASK	52.7	**0.0**	3.6	2.1	6.2	**0.0**	**3.1**
	VEIL	**62.0**	**0.0**	**0.5**	**0.5**	**1.0**	**0.0**	4.7
Caltech101	DP-FPL	74.3	**0.0**	3.1	1.0	**0.0**	1.0	**2.6**
	FedASK	60.6	**0.0**	3.1	2.1	2.1	1.6	6.8
	VEIL	**80.4**	1.0	**2.1**	**0.5**	1.0	**0.0**	3.1
DTD	DP-FPL	35.6	**0.0**	**1.0**	2.1	**0.0**	**0.0**	**0.5**
	FedASK	33.3	**0.0**	3.6	4.2	**0.0**	1.0	1.6
	VEIL	**37.6**	0.5	3.1	**1.0**	2.1	3.6	1.0

Table 3: Attack-specific comparison with private training mechanisms. All values are percentages averaged over three repetitions. Bold marks the best value within each dataset, including ties; shaded rows denote our method.

Variant	Acc. ↑	L	C	J	N	R	Q
Full VEIL	62.2	**0.0**	**0.0**	**0.0**	**3.1**	**0.0**	**0.0**
w/o S-I	62.1	**0.0**	**0.0**	**0.0**	**3.1**	1.6	**0.0**
w/o S-II	62.1	**0.0**	**0.0**	**0.0**	**3.1**	1.6	1.6
w/o S-III	**65.1**	**0.0**	**0.0**	**0.0**	6.2	1.6	3.1

Table 4: Stage-level ablation on Flowers102. All values are percentages. L, C, J, N, R, and Q denote FedMIA loss, cosine, joint, Nasr, RMIA, and Quantile MIA, respectively. Bold marks the best value including ties; the shaded row is the full method.

overall. The result supports distributional surrogate learning as a practical empirical complement to formally accounted mechanisms, not as a substitute for differential privacy.

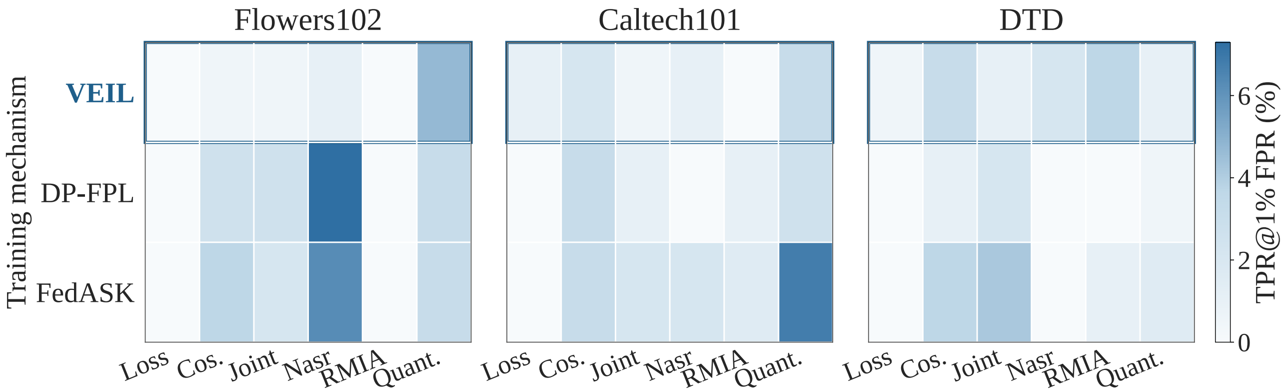

Figure 1: Attack surface of Veil, DP-FPL, and FedASK. Panels share one zero-anchored scale, darker cells indicate higher leakage, and the outlined row denotes our method. Numeric values are reported in Table 3.

References

Abadi, M.; Chu, A.; Goodfellow, I.; McMahan, H. B.; Mironov, I.; Talwar, K.; and Zhang, L. 2016. Deep Learning with Differential Privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*.

Bertran, M.; Tang, S.; Kearns, M.; Morgenstern, J.; Roth, A.; and Wu, Z. S. 2023. Scalable Membership Inference Attacks via Quantile Regression. In *Advances in Neural Information Processing Systems*.

Carlini, N.; Chien, S.; Nasr, M.; Song, S.; Terzis, A.; and Tramer, F. 2022. Membership Inference Attacks From First Principles. In *2022 IEEE Symposium on Security and Privacy*.

Chen, Z.; and Pattabiraman, K. 2024. Overconfidence Is a Dangerous Thing: Mitigating Membership Inference Attacks by Enforcing Less Confident Prediction. In *Network and Distributed System Security Symposium*.

Cimpoi, M.; Maji, S.; Kokkinos, I.; Mohamed, S.; and Vedaldi, A. 2014. Describing Textures in the Wild. In *2014 IEEE Conference on Computer Vision and Pattern Recognition*.

Duan, H.; Dziedzic, A.; Papernot, N.; and Boenisch, F. 2023. Flocks of Stochastic Parrots: Differentially Private Prompt Learning for Large Language Models. In *Advances in Neural Information Processing Systems*.

Fei-Fei, L.; Fergus, R.; and Perona, P. 2004. Learning Generative Visual Models from Few Training Examples: An Incremental Bayesian Approach Tested on 101 Object Categories. In *CVPR Workshop on Generative-Model Based Vision*.

Guo, T.; Guo, S.; Wang, J.; Tang, X.; and Xu, W. 2024. PromptFL: Let Federated Participants Cooperatively Learn Prompts Instead of Models—Federated Learning in the Age of Foundation Models. *IEEE Transactions on Mobile Computing*, 23(5).

Lester, B.; Al-Rfou, R.; and Constant, N. 2021. The Power of Scale for Parameter-Efficient Prompt Tuning. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*.

McMahan, B.; Moore, E.; Ramage, D.; Hampson, S.; and y Arcas, B. A. 2017. Communication-Efficient Learning of Deep Networks from Decentralized Data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*.

Nasr, M.; Shokri, R.; and Houmansadr, A. 2019. Comprehensive Privacy Analysis of Deep Learning: Passive and Active White-box Inference Attacks against Centralized and Federated Learning. In *2019 IEEE Symposium on Security and Privacy*.

Nilsback, M.-E.; and Zisserman, A. 2008. Automated Flower Classification over a Large Number of Classes. In *Sixth Indian Conference on Computer Vision, Graphics and Image Processing*.

Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; Krueger, G.; and Sutskever, I. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning*.

Shokri, R.; Stronati, M.; Song, C.; and Shmatikov, V. 2017. Membership Inference Attacks Against Machine Learning Models. In *2017 IEEE Symposium on Security and Privacy*.

Tran, L.; Sun, W.; Patterson, S.; and Milanova, A. 2025. Privacy-Preserving Personalized Federated Prompt Learning for Multimodal Large Language Models. In *International Conference on Learning Representations*.

Wen, M.; Zhu, J.; Xu, Y.; Zhou, Y.; and Han, D. 2025. Differentially Private Federated Low Rank Adaptation Beyond Fixed-Matrix. In *Advances in Neural Information Processing Systems*.

Zarifzadeh, S.; Liu, P.; and Shokri, R. 2024. Low-Cost High-Power Membership Inference Attacks. In *Proceedings of the 41st International Conference on Machine Learning*.

Zhao, H.; Du, W.; Li, F.; Li, P.; and Liu, G. 2023. FedPrompt: Communication-Efficient and Privacy-Preserving Prompt Tuning in Federated Learning. In *2023 IEEE International Conference on Acoustics, Speech and Signal Processing*, 1–5.

Zhou, K.; Yang, J.; Loy, C. C.; and Liu, Z. 2022. Learning to Prompt for Vision-Language Models. *International Journal of Computer Vision*, 130: 2337–2348.

Zhu, G.; Li, D.; Gu, H.; Yao, Y.; Fan, L.; and Han, Y. 2025. FedMIA: An Effective Membership Inference Attack Exploiting "All for One" Principle in Federated Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 20643–20653.

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2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) yes

2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) yes

2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) NA

2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA) NA

3. Dataset Usage

3.1. Does this paper rely on one or more datasets? (yes/no) yes

If yes, please address the following points:

3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA) yes

3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) NA

3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) NA

3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA) yes

3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA) yes

3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing (yes/partial/no/NA) NA

4. Computational Experiments

4.1. Does this paper include computational experiments?

(yes/no) yes

If yes, please address the following points:

4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) yes

4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) partial

4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) partial

4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) yes

4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) partial

4.7. If an algorithm depends on randomness, then its randomness controls are described in a way sufficient to allow replication of results (yes/partial/no/NA) yes

4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) yes

4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) yes

4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no) yes

4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no) yes

4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) no

4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) yes